

Weekly Status Report

Spectral Correlation with Event Cameras

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TL;DR The cross-color correlation is flat in effective frame rate ($10 \rightarrow 10,000$ FPS) and grows only with total averaging time T — confirming “it’s not the FPS, it’s the total duration.” The apparent FPS dependence in the earlier data was an artifact of how events were binned, now fixed by switching to the paper’s ON – OFF estimator, which also raises the converged correlation from 0.61 to 0.78. Single frames are detectable but are not reliable *estimators* of the spectral pattern; ≥ 5 averaged frames converge. The 60 Hz display caps scene dynamics, so the rotating-wheel experiment is the right next step — proposed protocol and predicted scaling inside.

Contents

1 Definitions — aligning on terms first

Symbol	Name	Definition
t	accumulation interval	wall-clock time integrated into one binned frame
$1/t$	effective frame rate (FPS)	binned frames produced per second
N	frame count	number of binned frames averaged
T	total averaging time	$T = N \cdot t$
r	cross-color Pearson r	correlation between the averaged frames of two wavelength recordings
c	noise ceiling	within-recording reliability (split-half, Spearman–Brown)

One correction to the proposed definition “effective frame rate = number of frames / accumulation time per frame”: with N frames each accumulated over t , that expression is N/t , which mixes units. The natural pair is

$$\text{FPS} = \frac{1}{t}, \quad N = \frac{T}{t},$$

so the two free knobs are (T, FPS) and N follows. This matters because the central finding is that **T is the operative variable and FPS is nearly free**. Under these definitions the TL;DR example from the thread — 1,000 frames at 10,000 FPS vs. 100 frames at 1,000 FPS, both $T = 100$ ms — verifies exactly: measured $r = -0.0043$ vs. -0.0045 (indistinguishable; both at the noise floor, since 100 ms is far below the ~ 0.5 s minimum), and at $T = 5$ s, $r = 0.7847$ vs. 0.7844 .

2 Responses to the open questions

Q1 — “Are you saying single events (very short durations) cannot be detected? That is exactly what the DVS is used for (e.g. lightning) and Ca spikes (the BU paper).”

*No — and this is the distinction I failed to make in the thread. **Detection and estimation are different problems.***

Every threshold crossing is detected and timestamped at microsecond resolution; nothing in our pipeline suppresses single events, which is exactly why DVS works for lightning and Ca transients.

Our task, however, is *estimation*: reconstructing the spatial diffraction pattern that encodes the spectrum from binned events. A single event, or a single binned frame, is one Poisson-distributed, threshold-quantized sample of that pattern. Quantitatively, in the paper’s event model a binned frame of window length t estimates the log-intensity change $\Delta \log I$ with error bounded by the contrast threshold C (their Supplement S4, Lemma S4.1):

$$\sum_{i=1}^n p_i = \frac{1}{C} (f(t_{\text{end}}) - f(t_{\text{start}})) + \varepsilon, \quad |\varepsilon| < 1,$$

and summing over N windows of a total period T with a scaling factor $a = C$:

$$\Delta \log I = a \cdot \sum_T \sum_t e(x, y, t, p) - \sum_{i=1}^N \varepsilon_i, \quad |\varepsilon_i| < 1.$$

The residuals ε_i of independent windows are roughly independent, so the N -frame average carries error $\sim C/\sqrt{N}$ where a single frame carries a worst-case C . This is the mechanism behind both our measured multi-frame floor and the paper’s Figure 2 (“large N nearly recovers L ”). Measured deviations of the reconstruction from the converged answer: **1–2 frames deviate by up to ~ 0.74 in correlation units; ≥ 5 averaged frames converge (≤ 0.075)**. The BU paper makes the same move from the tracking side: every frame is the sum of 16 refresh cycles, and their ablation without accumulation degrades tracking RMSE by +45% (neural phase mask) and +54% (neural amplitude mask).

Q2 — “Maybe you mean the spectral reconstruction error increases with effective frame rate (this is what I expect) and you have the data to show this.”

*Measured this week — the data shows **neither**. At fixed total averaging time, reconstruction quality is essentially **flat in effective frame rate**. What increases error is **less total averaging time**.*

Table 1: Mean cross-color Pearson r , ON – OFF estimator, at fixed total averaging time T vs. effective frame rate (LCD circle recordings).

T	10 FPS	1,000 FPS	10,000 FPS	FPS spread
5 s	0.7796	0.7844	0.7847	0.005
2 s	0.5675	0.5840	0.5847	0.017
1 s	0.3410	0.3605	0.3614	0.020
0.5 s	—	0.1582	0.1588	0.001

There *was* an apparent FPS dependence in the earlier data — up to 11.6% spread, in the direction you found counter-intuitive — but it was an artifact of folding ON and OFF events into one count before a nonlinearity. Binning with the paper’s signed scheme (ON – OFF, which their Supplement proves approximates $\Delta \log I$) collapses the spread to 0.5–2%, and it also *raises* the converged signal itself: $r(5\text{ s}) = 0.61 \rightarrow 0.78$, with green–red reaching 0.90. The 80% cross-color separation target — previously reported as unreachable — is now reached.

Q3 — “This is spectroscopy, there is no FoV; cross-correlation across the colors should be as low as possible. If it decreases with effective frame rate, does that mean colors differentiate better as you go faster? Counter-intuitive.”

Two clarifications: on FoV, and on the counter-intuitive reading.

1. **On “no FoV”:** agreed the deliverable is the spectrum / RGB value. The sensor’s pixel array is still the substrate that encodes it — the diffraction pattern is spatial, and the binned frame is the intermediate representation from which the spectrum is decoded. “Across the full aperture” in my earlier message meant: the estimator needs the event record covering the whole encoding pattern, not a spatial subset. The output remains a spectrum, not an image.
2. **On the counter-intuitive trend: your instinct was correct — it was a trap.** Cross-color r falling toward zero is only *better differentiation* if each recording is independently reliable. Below ~ 0.25 s of total averaging the recordings are not yet self-reproducible; the low r there is uncorrelated noise, not signal (as noted in the thread: “the lowest numbers are only just purely uncorrelated noise — not signal”). The correct statement is

differentiation = low cross-color r relative to a high noise ceiling c ,

and the ceiling c itself is built by total averaging time. Under the corrected estimator the FPS curve is flat — so no, FPS does not buy differentiation; total signal does.

Q4 — “Can you plot the cross-correlation as a function of the effective frame rate?”

Yes — Figure 1.

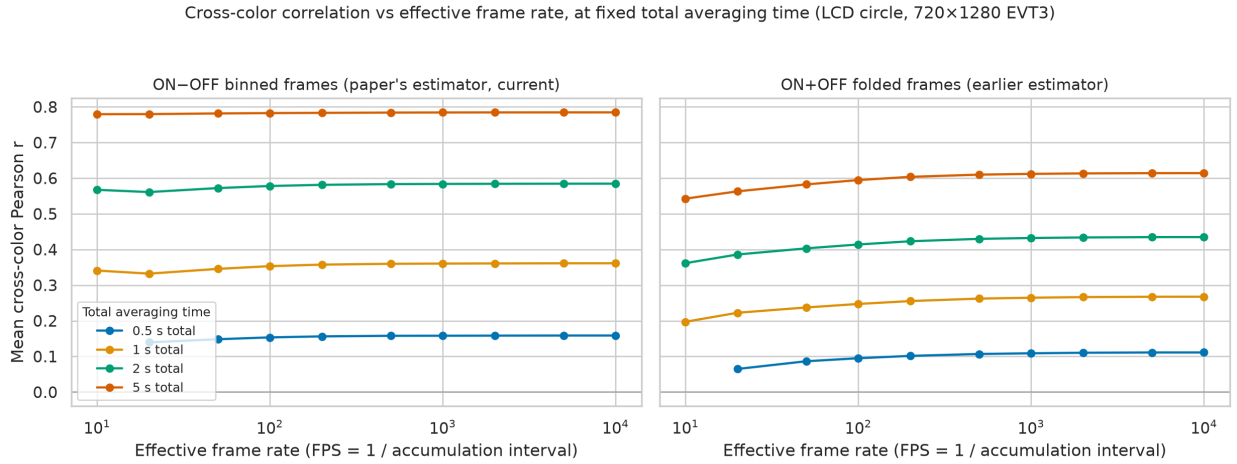


Figure 1: Mean cross-color Pearson r vs. effective frame rate at fixed total averaging times. **Left:** ON – OFF binned frames (paper’s estimator, current). **Right:** earlier ON + OFF folded estimator — this is where the apparent FPS trend lived, and the panel makes visible that it was an artifact. Both panels share the y -axis; each curve is a fixed total averaging time (0.5/1/2/5 s); the x -axis spans effective FPS from 10 to 10,000.

Q5 — “Oh so you mean the display is not bright enough?”

No — *brightness is not the limiter. The display is a 60 Hz screen: content updates every 16.7 ms, and the backlight/refresh modulates the sensor even for static content.*

Measured consequences this week:

- The first accumulation interval containing reproducible structure is ~ 16 ms — **one display period** — which we now treat as a display-period artifact, not a fundamental limit. (The BU paper’s “16 frames at 1 ms” coincidence is worth re-examining on hardware that is not refresh-locked.)
- Split-half reliability diagnostics go negative in a pattern consistent with windows sampling fixed phases of a ~ 60 Hz-periodic modulation.
- Practical upshot: the *scene’s* temporal dynamics are capped by the display, so the “how fast can the scene change” axis cannot be probed on this source — hence the rotating-wheel experiment below.

Q6 — “With a faster source you can probe the limit I am looking for, right? Rotating filter wheel with a lamp or bright source; add color filters for different spectra?”

Agreed — that is the right next experiment, and it directly tests the claim that smaller accumulation intervals require more dynamic signal. One design upgrade to consider.

Rather than a continuously rotating *filter* wheel, the already-CAD’d 18-LED chopper wheel (63 mm, 20° station spacing, `models/`) puts the timing under precise control: mounting two wavelength LEDs in the wheel gives a known, stable modulation frequency *and* the spectral pair in the same instrument, with the same correlation pipeline as output.

The predicted scaling follows from the event-rate physics: a pixel fires when the log-intensity change accumulates past the contrast threshold C , so a window of length t captures roughly

$$n_{\text{events}} \approx \frac{t |d \log I / dt|}{C}$$

events per active pixel. Enough events per window requires

$$t_{\min} \approx \text{a few} \cdot \frac{C}{|d \log I / dt|} \implies t_{\min} \propto \frac{1}{f},$$

i.e. **the minimum usable accumulation interval scales inversely with the modulation frequency**. Equivalently: the governing quantity is **events per window**, not FPS.

Protocol:

1. Frequency ladder incommensurate with 60 Hz (e.g., 97, 211, 503 Hz) + one deliberate 60 Hz-commensurate run as a positive control for aliasing.
2. Chopper-off baseline capture: characterizes the display’s intrinsic modulation and provides the event-starved (“static scene”) end point, where $t_{\min}$ should blow up.
3. For each frequency: 10 s recording, full sweep with ON – OFF binning; extract events-per-window, minimum t with significant r , and the FPS-invariance check.

If the $t_{\min} \propto 1/f$ scaling holds — equivalently, if results collapse onto a single curve in events-per-window — that confirms “smaller t requires faster dynamics” quantitatively and gives the t -vs-frequency design map for the spectroscopy experiments.

Q7 — “If you need more frames, we can define the effective frame rate accordingly: number of frames / accumulation time per frame.”

Agreed on precise definitions — proposal in §1: $\text{FPS} = 1/t$, $T = N \cdot t$, $N = T \cdot \text{FPS}$. Under these, the TL;DR example verifies exactly (numbers in §1).

3 Accomplishments this week

- Completed the fine duration $\times$ frame-rate sweep instrument: 12–25 ms resolution grid, ON – OFF / ON / OFF / folded polarity modes, split-half noise ceiling, per-pair Student- t significance.
- Switched the estimator to the BU paper’s ON – OFF binning (proven $\approx \Delta \log I$, their Supplement S4): converged cross-color correlation $0.61 \rightarrow 0.78$, green–red 0.90; the 80% separation target is now reached.
- Diagnosed the earlier FPS-dependent trend as a polarity-folding artifact; FPS spread collapsed from 11.6% to 0.5–2%.
- Verified all four shared claims against Shah et al. (CVPR 2024), including their 16-refresh-cycle binning and their no-accumulation ablation (+45–54% RMSE) matching our multi-frame requirement.
- Produced Figure 1 (the requested cross-correlation vs. effective-frame-rate plot); verification report deck delivered (`results/polarity_verification_report.pptx`).

4 In progress

Task	Status	Expected completion
Chopper LED wheel mounting + frequency ladder captures	hardware ready, awaiting bench time	next week
Partition-invariance test (ON – OFF, linear scaling; no new capture needed)	queued	early next week
Sub-10-frame floor re-measurement under the corrected estimator	queued	next week

5 Blockers

60 Hz display ceiling — the current source cannot probe scene dynamics above one refresh period, and it injects its own periodic modulation (the likely cause of a negative reliability diagnostic in the split-half analysis). Impact: the $t_{\min}$ -vs-frequency scaling cannot be measured on the LCD. Resolution: the chopper/LED wheel captures above — hardware exists, needs mounting.

6 Next week’s priorities

1. Chopper bench setup; frequency ladder captures (incommensurate with 60 Hz + one commensurate control + chopper-off baseline).
2. Partition-invariance test on existing recordings (validates the estimator before new captures are analyzed).
3. $t_{\min} \propto 1/f$ scaling analysis from the frequency ladder; updated FPS plot with chopper data.
4. Revised spectra-deck conclusions (80% target reached; 16 ms finding re-scoped as a display-period artifact pending chopper data).

7 Standing operating envelope (from this week’s measurements)

≥ 5 averaged frames; $\geq 0.5\text{--}1$ s total averaging for reliable discrimination; effective FPS free from 10 to 10,000.

The paper’s 16 ms bin sits at the lower *edge* of detectability on display-driven content; treat it as an edge, not an operating point, until the chopper data re-baselines it.

Data: LCD circle recordings, 720×1280 EVT3, 5 s per color; crop + spatial high-pass + support-masked Pearson; ON – OFF binning per Shah et al. Supplement S4. Scripts and result tables in `src/hypercam/duration_sweep.py` and `src/hypercam/results/`.